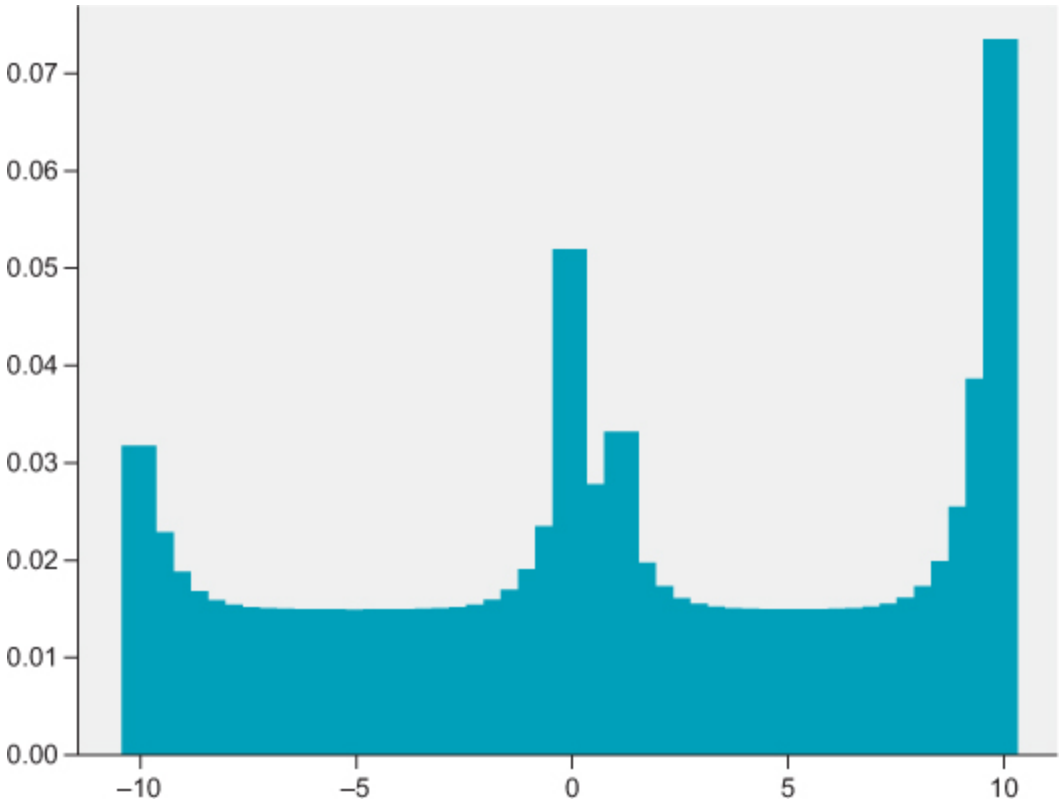


Now let’s see how the variance decreases if we observe the same reward multiple times, starting from a uniform prior.

Listing 7.5. Decreased variance with sequence of same reward

```
ob_rewards = [5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5,
5, 5]
for i in range(len(ob_rewards)):
    Z = update_dist(ob_rewards[i],
torch.from_numpy(support).float(), \
                            Z, lim=(vmin,vmax), gamma=0.7)
plt.bar(support, Z)
```

You can see in figure 7.14 that the uniform distribution transforms into a normal-like distribution centered at 5 with much lower variance. We will use this function to generate the target distribution that we want the Dist-DQN to learn to approximate. Let’s build the Dist-DQN now.

Figure 7.14. The result of updating an initially uniform probability distribution after observing the same reward multiple times. The uniform distribution converges toward a normal-like distribution.